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A APPENDIX

A.1 SPARSE AUTOENCODER TRAINING

A.1.1 TRAINING DATASET

We train our model using the Astral SCOPe 2.08 dataset, filtered to 40% sequence identity, which includes approximately 15k highly non-redundant protein sequences (Fox et al., 2014). This dataset provides a manageable number of tokens, enabling faster iteration over different hyperparameters while maintaining a diverse range of protein sequences and structural domains. This diversity allows the autoencoder to learn a broad spectrum of features.

A.1.2 HANDLING OF DEAD LATENTS

We check how frequently each of the latents activates over a subsample of tokens (50 batches of 4096 tokens) at regular intervals during training (every 500 batches). When this frequency is close to zero ($< 10^{-5}$), we consider that the latent is “dead” and we re-initialize its weights to “revive” it.

A.1.3 EVALUATION METRICS

To decide which hyperparameters (learning rate, sparsity penalty λ , SAE hidden size n) produce the best sparse autoencoder, we use the following metrics:

- L_0 : The average number of non-zero components in the latent vector z for a given amino acid token. This is our measure of the sparsity level of the autoencoder.
- Number of dead latents: The number of components in the latent space that are never non-zero over a large number of sample tokens ($\sim 10^5$). This is a general metric for sparse autoencoders quality.
- Cross Entropy (CE) Increase: Difference between the average cross entropy loss of the original model and the cross entropy loss of the model when we substitute the activations in a given layer by the corresponding activations reconstructed by the autoencoder. This indicates how much of the model’s performance the sparse autoencoder fails to reconstruct.

A.1.4 HYPERPARAMETER SELECTION

We perform a hyperparameter sweep with the following values: learning rate [$5e^{-4}$, $1e^{-4}$, $1e^{-3}$]; L_1 penalty [0.0003, 0.001, 0.005] and dictionary size multiplier [32, 10, 5].

We use the evaluation metrics detailed in section A.1.3 to decide on the best combination of hyperparameters. Specifically, we aim to find a model that balances reconstruction error and sparsity by focusing on the “elbow” part of the CE increase against L_0 plot, where increasing the density of active latents in the latent space does not significantly reduce the CE increase, indicating an optimal trade-off.

A.2 INTERPRETABILITY

A.2.1 FEATURE ANNOTATION DATA

As ground truth features for the interpretability analysis, we use the following protein annotations from Uniprot version 2024.1 :

- Transmembrane region
- Topological domain
- Binding site
- Zinc finger region
- Region of interest
- Intramembrane region
- Active site
- Disulfide bond
- Glycosylation site
- Helix
- Turn
- Strand

A.3 ADDITIONAL RESULTS

A.3.1 INTRINSIC DIMENSION ANALYSIS

We estimate the intrinsic dimension across all layers of ESM-2 8M following the methodology outlined by (Valeriani et al., 2024). The resulting curve, shown in Figure 4, guides our decision to extract embeddings from layer 3. Larger models in the ESM-2 family exhibit a similar intrinsic dimension profile across different model sizes, with a more pronounced plateau region of comparatively low intrinsic dimension in the same relative layer position.